

Twitter Emotion Analysis - Project Documentation

Overview

Twitter Emotion Analysis is a Natural Language Processing (NLP) project that analyzes emotions expressed in tweets based on a user-provided hashtag or keyword. The system fetches tweets using the Twitter API via Tweepy and performs emotion analysis using TextBlob.

Features

- Analyze emotions from tweets using NLP.
- Fetch real-time tweets based on hashtags or keywords.
- Generate detailed emotion analysis reports.
- Display emotion distribution using charts.
- Export analyzed data to CSV files.
- Supports large tweet datasets.
- Tested with 100+ keywords and 5000+ tweets.
- Approximately 92% accuracy.

Technologies Used

Python, Tweepy, TextBlob, Pandas, NumPy, Matplotlib, CSV

Installation

1. Clone the repository.
2. Create and activate a virtual environment.
3. Install dependencies using `pip install -r requirements.txt`.
4. Configure Twitter API credentials.
5. Run the application using `python main.py`.

Workflow

1. User enters a hashtag or keyword.
2. Tweets are fetched from Twitter.
3. Text preprocessing is performed.
4. Emotion analysis is executed using NLP techniques.
5. Results are visualized through charts.
6. Data is exported to CSV.

Future Enhancements

- Deep Learning based emotion classification.
- Real-time dashboard.
- Multi-language support.
- BERT-based emotion analysis.
- Flask/Django web deployment.

Author

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